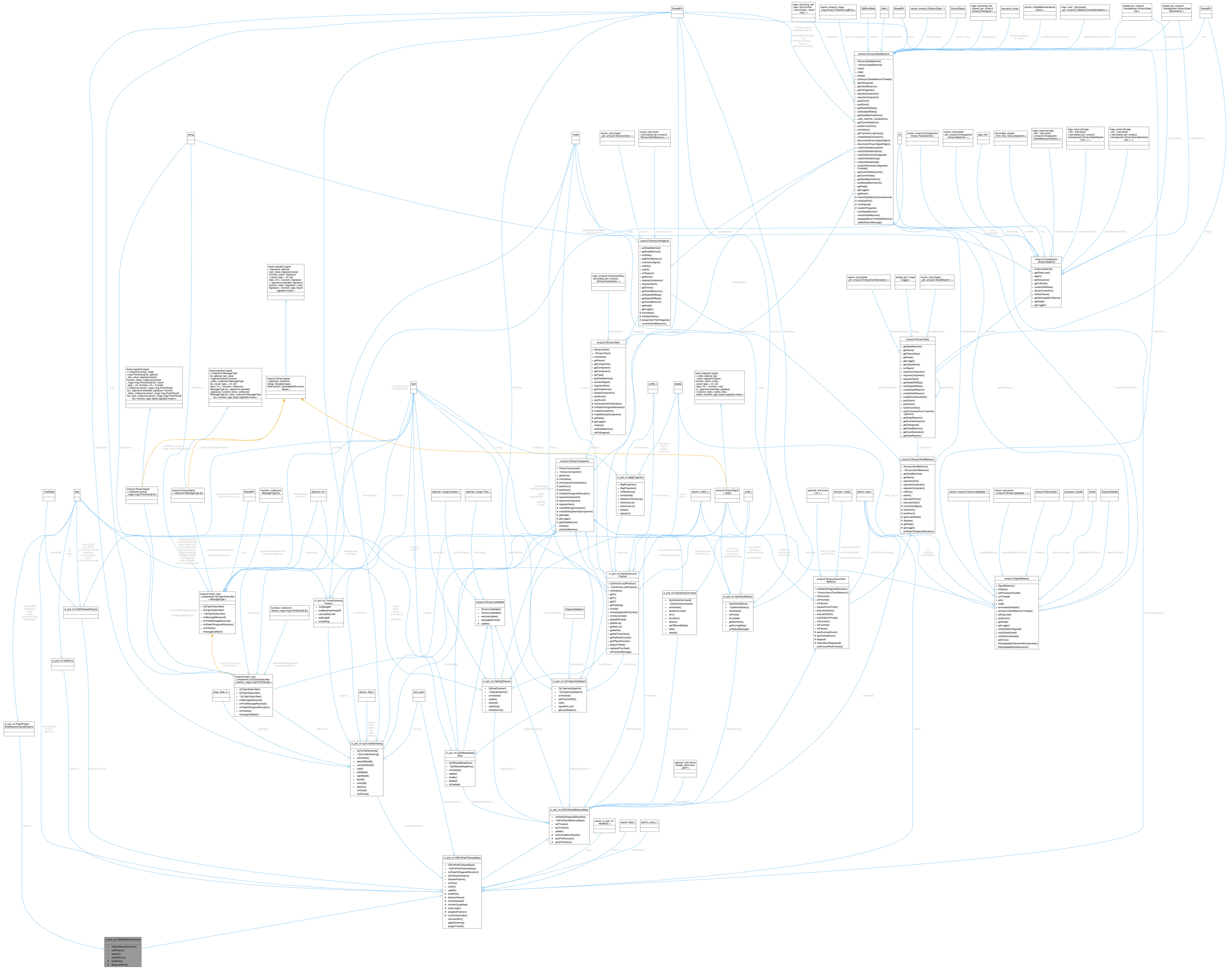
cl_px4_mr::CbSineWaveHorizontal
+ CbSineWaveHorizontal()
+ setParams()
+ params()
+ setEndPoint()
# buildPath()
# behaviorName()
cl_px4_mr::CbPx4PathFollowerBase
+ CbPx4PathFollowerBase()
+ ~CbPx4PathFollowerBase()
+ onStateOrthogonalAllocation()
+ setFollowerParams()
+ followerParams()
+ onEntry()
+ onExit()
+ update()
# buildPath()
# behaviorName()
# onPathStarted()
# onPathCompleted()
# totalLength()
# progressFraction()
# carrotVertexIndex()
- commandFor()
- applyCentering()
- tangentYawAt()
cl_px4_mr::CbPx4ClientBehaviorBase
+ onStateOrthogonalAllocation()
+ ~CbPx4ClientBehaviorBase()
+ setTimeout()
+ hasTimeout()
+ update()
# wireCompletionSignals()
# postPx4Success()
# postPx4Failure()
vector< cl_px4_mr::NedPoint >
vector< float >
atomic< size_t >
-path_
-cumLen_
-carrotVertex_
-watchdogStart_
optional< std::chrono::steady_clock::time_point >
#offboardKeepAlive_
-lastUpdateTime_
#goalChecker_
#trajectorySetpoint_
#localPosition_
#vehicleCommand_
#tunnelCentering_
cl_px4_mr::CpOffboardKeepAlive
+ CpOffboardKeepAlive()
+ ~CpOffboardKeepAlive()
+ onInitialize()
+ update()
+ enable()
+ disable()
+ isEnabled()
cl_px4_mr::CpGoalChecker
+ CpGoalChecker()
+ ~CpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()
cl_px4_mr::CpTrajectorySetpoint
+ CpTrajectorySetpoint()
+ ~CpTrajectorySetpoint()
+ onInitialize()
+ setPositionNED()
+ hold()
+ republishLast()
+ getLastSetpoint()
TrajectorySetpoint
-lastSetpoint_
-localPosition_
-trajectorySetpoint_
+onGoalReached_
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
-goalActive_
-periodDuration_
-lastUpdate_
-hasPublished_
-enabled_
-lastCloudNs_
-lastLog_
time_point
cl_px4_mr::CpVehicleLocalPosition
+ CpVehicleLocalPosition()
+ ~CpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ isHeadingGoodForControl()
+ isVelocityValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlt()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
- onPositionMessage()
-projection_
+onLocalPositionReset_
+onPositionReceived_
-xyResetCounter_
-zResetCounter_
-timeoutMs_
cl_px4_mr::CpVehicleCommand
+ CpVehicleCommand()
+ ~CpVehicleCommand()
+ onInitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()
cl_px4_mr::CpTunnelCentering
+ CpTunnelCentering()
+ ~CpTunnelCentering()
+ onInitialize()
+ lateralOffsetM()
+ verticalOffsetM()
+ valid()
+ leftWallM()
+ rightWallM()
+ floorM()
+ ceilingM()
+ params()
- onCloud()
- onAttitude()
-ceiling_
-floor_
-lateral_
-left_
-right_
-vertical_
atomic< float >
-params_
-haveAttitude_
-attitudeQ_
-subscriber_
array< float, 4 >
cl_px4_mr::TunnelCenteringParams
+ minRangeM
+ wallBandHalfHeightM
+ verticalSearchM
+ wallFadeM
+ smoothing
function< void(const sensor_msgs::msg::PointCloud2 &)>
smacc2::client_core_components::CpTopicSubscriber< sensor_msgs::msg::PointCloud2 >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
< sensor_msgs::msg::PointCloud2 >
+postInitialMessageEvent
+postMessageEvent
+queueSize
smacc2::client_core_components::CpTopicSubscriber< MessageType >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ onInitialize()
- messageCallback()
+onFirstMessageReceived_
+onMessageReceived_
-sub_
-firstMessage_
-initialized_
-goalX_
-goalY_
-goalZ_
-xyTolerance_
-zTolerance_
-heading_
-refAlt_
-x_
-y_
-z_
+ceilingClearanceM
+columnHalfWidthM
+floorClearanceM
+floorLookBackM
+lateralSearchM
+lookAheadMaxM
+lookAheadMinM
+maxLateralM
+maxVerticalM
+minFloorClearanceM
...
cl_px4_mr::PathFollowerParams
cl_px4_mr::NedPoint
cl_px4_mr::FlightPattern::SineWaveHorizontalParams
YawMode
float
smacc2::SmaccSignal< void(const sensor_msgs::msg::PointCloud2 &)>
smacc2::SmaccSignal< void(const MessageType &)>
SharedPtr
function< void(const MessageType &)>
optional< int >
smacc2::SmaccSignal< Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal
string
bool
mutex
smacc2::ISmaccComponent
smacc2::ISmaccClient
smacc2::ISmaccOrthogonal
smacc2::ISmaccStateMachine
smacc2::ISmaccState
smacc2::ISmaccClientBehavior
smacc2::SmaccAsyncClientBehavior
smacc2::SignalDetector
smacc2::introspection::SmaccStateInfo
cl_px4_mr::MapProjection
cl_px4_mr::CpVehicleStatus
uint64_t
double
uint8_t
atomic< int8_t >
smacc2::SmaccSignal< void()>
optional< rclcpp::Duration >
optional< rclcpp::Time >